ABM Staging: Turn On and Off

Project `abmgenerator`, region us-central1. Staging: https://136-81-60-29.sslip.io

What this is

Staging costs money while it is idle, so we switch it off when nobody is testing and back on when we need it. Two scripts do this from a Mac:

- `abm-offline.sh` switches staging off.
- `abm-online.sh` switches staging back on.

Nothing is deleted. Switching off only stops the parts that bill by the hour, and switching on starts them again.

One-time setup on your Mac

1. Install the Google Cloud CLI (gcloud):

   ```
   brew install --cask google-cloud-sdk
   ```

2. Get the two scripts from the project folder: `scripts/mac/abm-offline.sh` and `scripts/mac/abm-online.sh`.
3. The scripts sign you in on first run. Use the Google account that has access to project `abmgenerator`. A browser window opens for the sign-in.

How to run

From the project folder, in Terminal:

To switch staging off:

```
bash "scripts/mac/abm-offline.sh"
```

You will be asked to type `yes` before anything changes. Anything else cancels.

To switch staging on:

```
bash "scripts/mac/abm-online.sh"
```

Switching on waits for the database to start, which can take a few minutes. Each step prints what it did.

What each script changes

Step	Switch off (abm-offline.sh)	Switch on (abm-online.sh)
1	Turns off the enabled alert policies, so the stop does not send alerts.	Starts the database `ai-staging-pg` and waits until it is ready.
2	Cloud Run `ai-staging-web`: public traffic off (ingress internal), minimum instances 0. Cloud Run `ai-staging-api`: minimum instances 0.	Cloud Run `ai-staging-api`: minimum instances 1. Cloud Run `ai-staging-web`: public traffic back on (load balancer), minimum instances 1.
3	Pauses 9 scheduler jobs (schedules-tick, isolation-check, access-expiry, audit-verify, metrics, maildrive-sync, notify-relay, spend-invariants, sweep).	Resumes the same 9 scheduler jobs.
4	Pauses 3 task queues (runs, jobs, documents).	Resumes the same 3 task queues.
5	Stops the database `ai-staging-pg`.	Turns the alert policies back on. The "5xx errors" and "latency p95" alerts stay off on purpose.

Switching off goes alerts first, then Cloud Run, scheduler jobs and queues, then the database. Switching on goes database first, then Cloud Run, queues and jobs, then alerts. The Cloud Run worker is not changed: it already has a minimum of 0 and only runs when tasks arrive. The order matters: the database has to be running before anything that uses it. Migrations fail with a connection timeout while it is stopped.

What still costs money while staging is off

- Redis (`ai-staging-rate`), about $35 a month.
- External load balancer with Cloud Armor, about $25 a month. Removing it would change the staging address and break Google sign-in, so the scripts leave it alone.
- Cloud NAT, about $5 a month.
- Artifact Registry images, about $3 a month, and the stopped database's storage.

Cloud Run worker scales to zero when idle. The API and web keep 1 warm instance each while staging is on, so they cost money until the off script sets their minimum to 0.

Checking the state

- After switching on, `https://136-81-60-29.sslip.io/readyz` should return 200.
- Opening `https://136-81-60-29.sslip.io/v1/models` should return 401. That means the API is up and asking for sign-in.

Troubleshooting

What you see	What to do

`gcloud is not installed`	Run `brew install --cask google-cloud-sdk`, then run the script again.
Permission denied, or the project is not found	You are signed in with the wrong Google account. Run `gcloud auth login` and sign in with the account that has access to `abmgenerator`.
A step says "could not resume" or "could not pause"	Usually the item is already in that state. Check it in the Cloud Console before retrying.
The alerts step says alpha is missing	Run `gcloud components install alpha`. The other steps do not need it.
Database still not ready after the wait	Run `gcloud sql instances describe ai-staging-pg --project abmgenerator`. If it says RUNNABLE, run the script again. If not, ask the owner.
Migrations fail right after switching on	The database was not ready. Wait until it shows RUNNABLE, then run the migration job again.

> **Before you switch staging off, check that nobody is testing.** Ask in the team channel first.